

CO4-AT2-Visualization Evaluation Exercise

Name: A Roja

Reg No: 192424226

Course: DSA0601-Data Handling and Visualization

Code of Conduct:

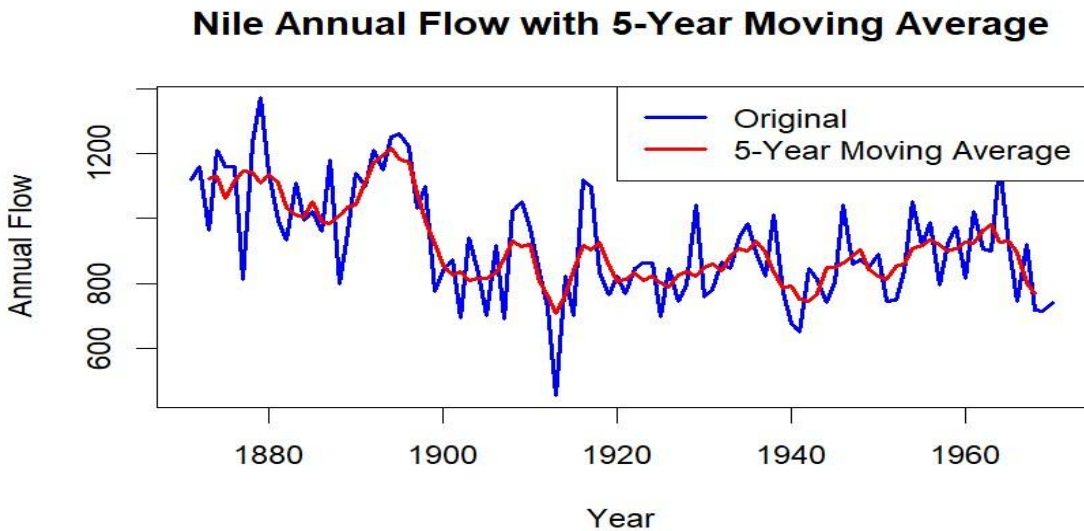
I A Roja [192424226] (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: A Roja.

Question 1: Individual Time Series & Moving-Average Smoothing

Code:

```
nile_ts <- ts(Nile, start = 1871, frequency = 1) ma5 <-  
filter(nile_ts, rep(1/5, 5), sides = 2) plot(nile_ts, type = "l",  
col = "blue", lwd = 2, xlab = "Year", ylab =  
"Annual Flow", main = "Nile Annual Flow with 5-Year  
Moving Average") lines(ma5, col = "red", lwd = 2)  
legend("topright", legend = c("Original", "5-Year  
Moving Average"), col = c("blue", "red"), lty = 1,  
lwd = 2)
```



Observation

The Nile series shows a noticeable change in level around the late 1890s to early 1900s, where the annual flow appears to decrease. The moving-average line makes this shift clearer by reducing short-term fluctuations. The variability also appears different before and after the level change.

Question 2: Multiple Time Series Comparison & Dose–Response Curve

Part A: EuStockMarkets Dataset

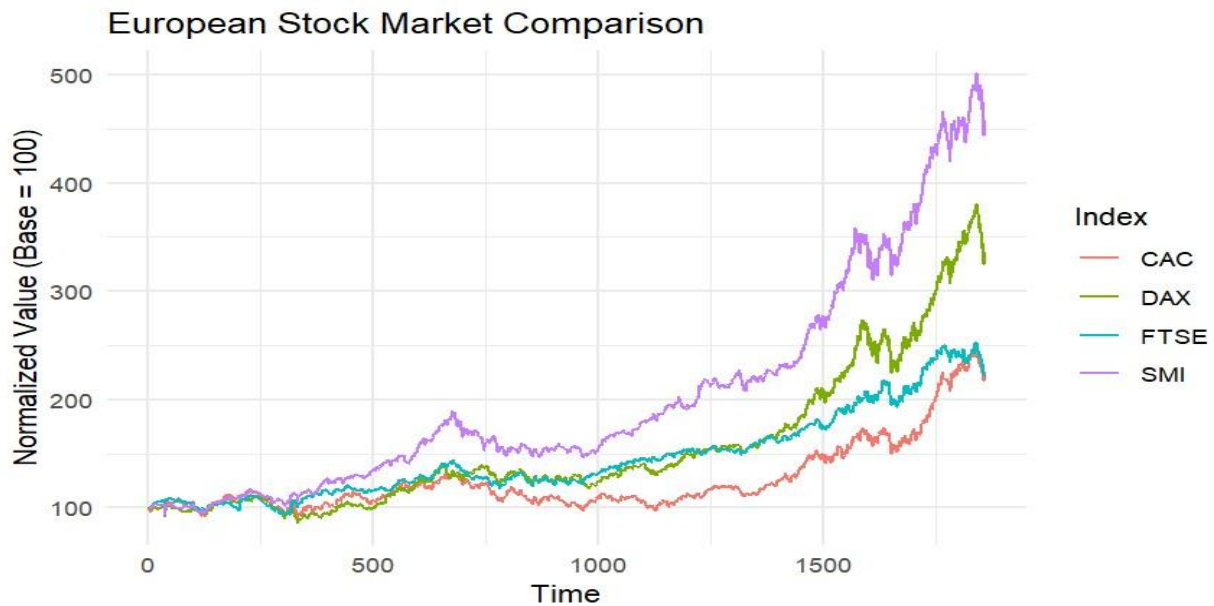
Code:

```
library(ggplot2)
library(tidyr) library(dplyr) stock_data <-
as.data.frame(EuStockMarkets) stock_data$Time <-
1:nrow(stock_data) stock_long <- stock_data %>%
pivot_longer( cols = c(DAX, SMI, CAC, FTSE),
names_to = "Index", values_to = "Value")
stock_normalized <- stock_long %>%
group_by(Index) %>% mutate(Normalized_Value =
Value / first(Value) * 100) ggplot(stock_normalized,
```

```

aes(x = Time,      y = Normalized_Value,
color = Index)) + geom_line() + labs(  title =
"European Stock Market Comparison",  x = "Time",
y = "Normalized Value (Base = 100)" ) +
theme_minimal()

```



Observation

The index whose normalized line finishes at the highest value shows the strongest overall growth during the period. After re-indexing all series to 100, their relative percentage growth can be compared even though their original values are on different scales

Part B: Ryegrass Dose-Response

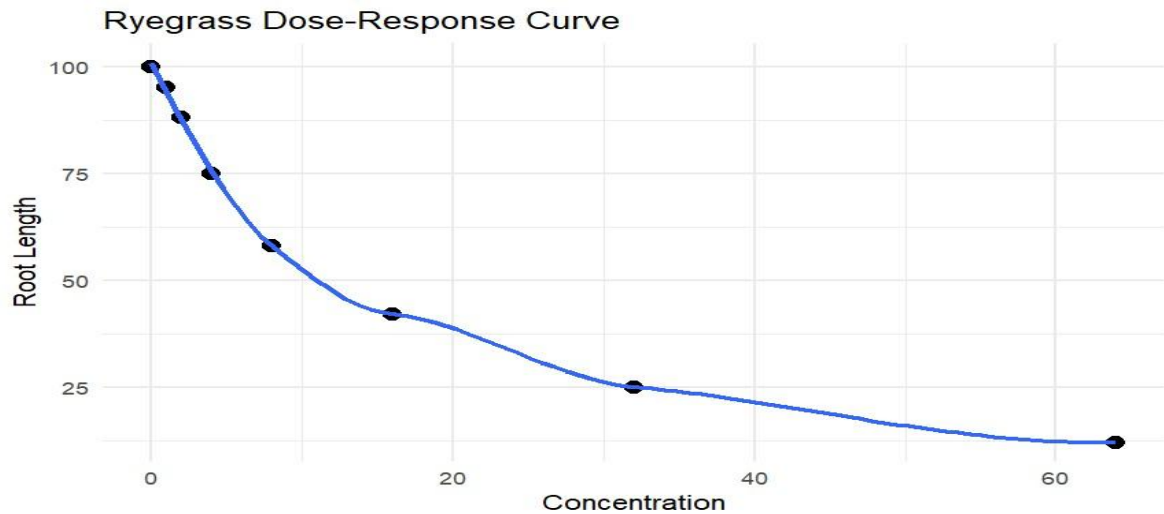
Code:

```

library(ggplot2) ryegrass <- data.frame(
concentration = c(0, 1, 2, 4, 8, 16, 32, 64),
root_length = c(100, 95, 88, 75, 58, 42, 25, 12))
print(ryegrass) ggplot(ryegrass, aes(x =
concentration,      y = root_length)) +
geom_point(size = 3) + geom_smooth(method

```

```
= "loess", se = FALSE) + labs( title =
"Ryegrass Dose-Response Curve", x =
"Concentration", y = "Root Length" ) +
theme_minimal()
```



Observation

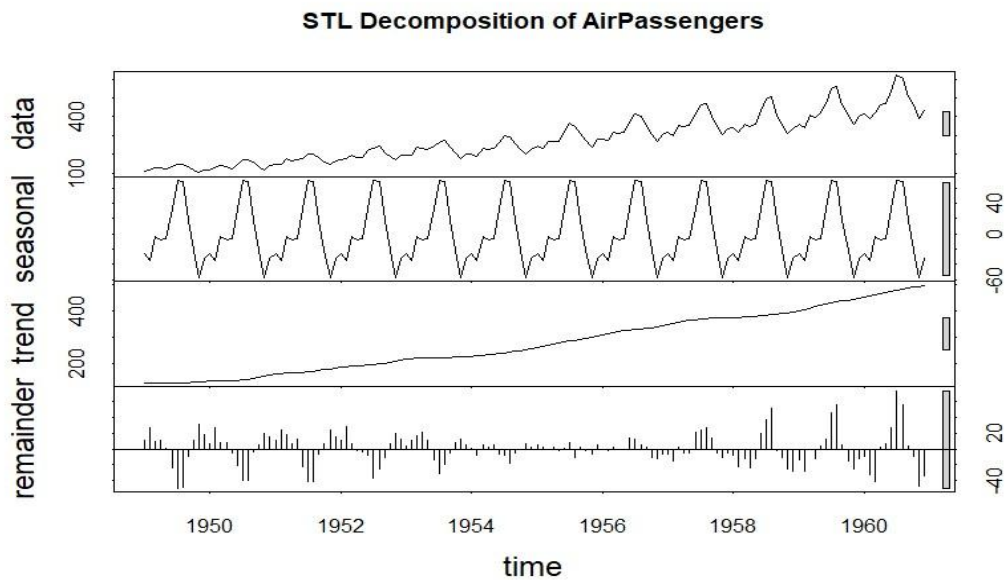
As concentration increases, root length generally decreases, producing a negative dose-response relationship. The concentration returned in **result** is the approximate concentration at which root length is reduced to about 50% of the control value.

Question 3: STL Decomposition of AirPassengers

Code:

```
air_ts <- ts(
  AirPassengers, start
  = c(1949, 1),
  frequency = 12)
air_stl <- stl( air_ts,
```

```
s.window = "periodic") plot(air_stl, main =  
"STL Decomposition of AirPassengers")
```



Observation

The AirPassengers data shows an increasing trend over time along with strong yearly seasonal patterns. The seasonal fluctuations become larger as the number of passengers increases, so a **multiplicative decomposition** is more appropriate than a purely additive decomposition.

Question 4: Detrending and Residual Analysis of CO₂

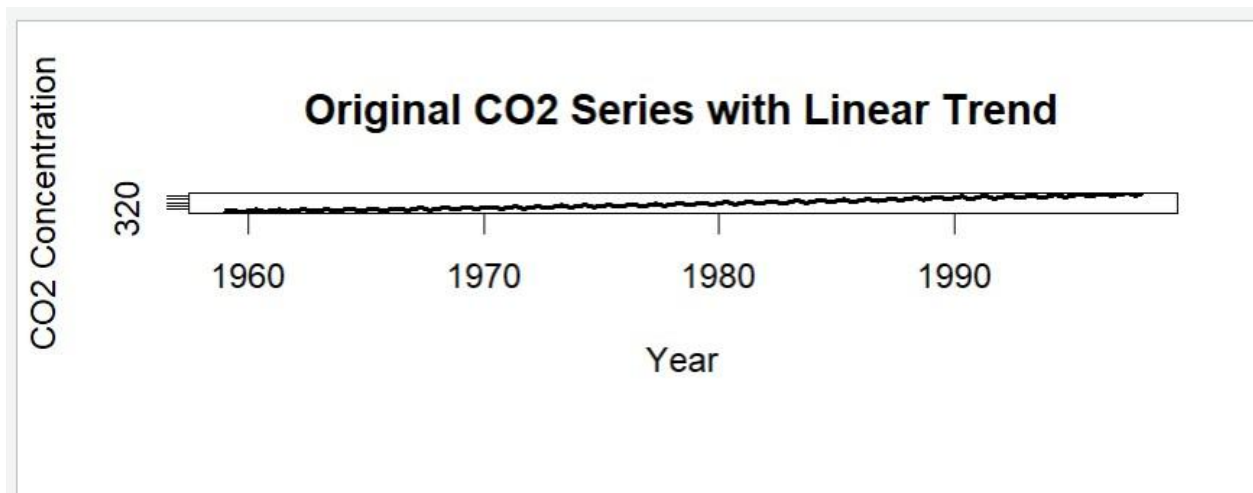
Code:

```
data(co2) co2_values <- na.omit(co2)
time_values <- time(co2_values)
co2_numeric <- as.numeric(co2_values)
model <- lm(co2_numeric ~ time_values)
trend <- fitted(model)
detrended <- residuals(model) par(mfrow = c(2, 1))
plot(time_values, co2_numeric, type = "l",
xlab = "Year", ylab = "CO2 Concentration",
```

```

main = "Original CO2 Series with Linear Trend",
lwd = 2) lines(time_values,
              trend, lwd = 2, col = "red")
legend("topleft", legend = c("Original
CO2", "Linear Trend"), col = c("black",
"red"), lty = 1, lwd = 2)
plot(time_values, detrended, type = "l",
xlab = "Year", ylab = "Residual",
      main = "Detrended CO2 Residuals",
lwd = 2) abline(h = 0, col = "red",
lty = 2)
par(mfrow = c(1, 1))

```



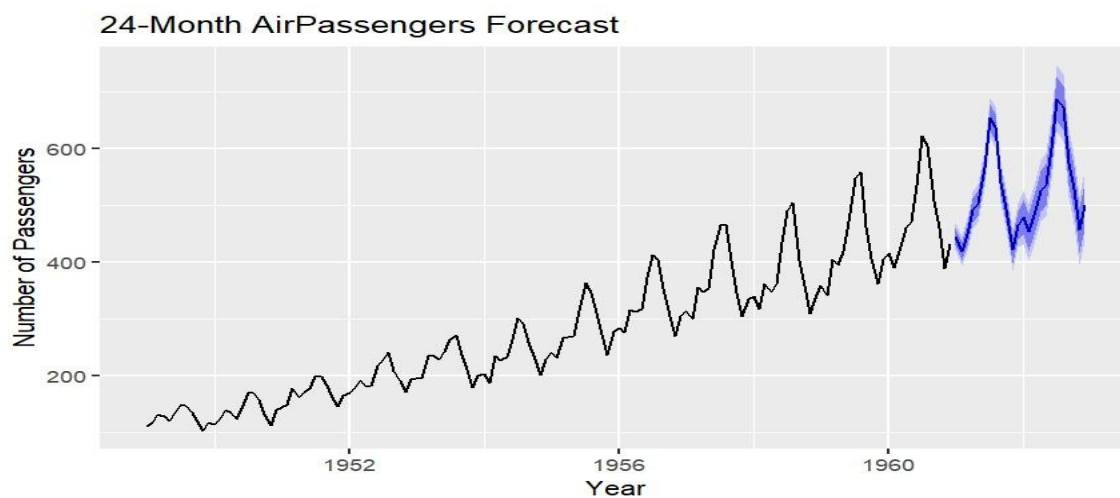
Observation

After removing the linear trend, a clear repeating seasonal pattern remains in the residuals. Therefore, simple linear detrending is not sufficient because the CO₂ series contains both a long-term upward trend and strong seasonal variation.

Question 5 – Part A: Forecasting AirPassengers

CODE:

```
install.packages("forecast")
library(forecast) air_ts <-
AirPassengers model <-
auto.arima(air_ts)
forecast_result <- forecast(
model, h = 24)
autoplot(forecast_result) +
ggtitle("24-Month
AirPassengers Forecast") +
xlab("Year") +
ylab("Number of
Passengers")
```



Observation

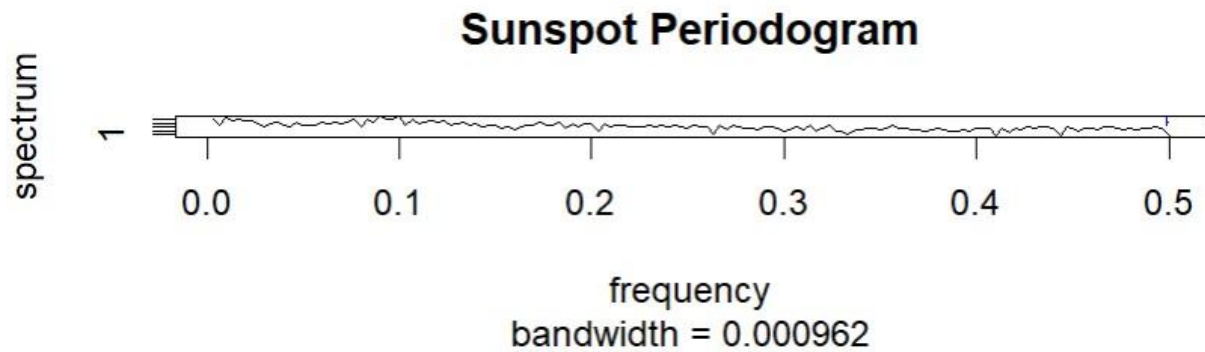
The forecast continues the repeating yearly seasonal pattern observed in the historical AirPassengers data. The peaks and low points occur in a similar cycle, indicating that the model has captured the seasonality reasonably well.

Part B: Sunspot Cycle Detection

CODE:

```
sunspot_ts <- sunspot.year acf(  
sunspot_ts, main = "ACF of  
Sunspot Data" )
```

```
spectrum( sunspot_ts, main =  
"Sunspot Periodogram"  
)
```



Observation

The ACF and periodogram typically indicate a repeating cycle of approximately **10–11 years**. This is consistent with the commonly cited solar cycle of approximately **11 years**, although the exact cycle length can vary slightly.